
AberdeenGroup

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Marathon's *Endurance*TM 4000: Bringing Fault Tolerance to NT Servers & Clusters

New enterprise applications are increasingly targeted for lower-cost hardware platforms running Microsoft's NT Server operating system. Marathon's Endurance 4000 brings fault tolerance to mission-critical NT applications.

Caution Ahead

Aberdeen research shows an ever-increasing trend towards using the NT Server® operating system for a wide variety of enterprise applications, including transaction processing, electronic mail and groupware, and internet and intranet sites. Moreover, NT is also being used for network management activities such as hosting software-based firewalls.

Here's the caution: NT Server was never designed to run mission-critical applications without failing. No general purpose operating system makes that claim. So, as more and more of the enterprise application jewels are hosted by NT servers, the risk of significant business disruption from an NT server failure rises exponentially as these servers are deployed like popcorn. What's a savvy but paranoid Information Technology executive to do?

Microsoft's Wolfpack is *an* Answer, But Not *The* Answer

Wolfpack is the widely discussed software from Microsoft that will provide NT server-failover clustering when it is delivered later in 1997. Wolf-pack technology will restart an application on an attached cluster partner. But please note that the recovery process will be measured in minutes as a reboot is required.

While "minutes to recover" will be good enough for many applications, it should not be considered acceptable for truly mission-critical applications where business and customer interactions are most important. We would put almost all OLTP, system management, and firewall applications in the critical category — but also many e-mail, file serving and groupware applications.

But to date, IT buyers could not really protect their mission-critical servers, only wait for Wolfpack to be delivered, or use one of the cluster-software middleware solutions from suppliers including DEC, NCR, and Tandem. However, most cluster software solutions require special application versions to ride through a failure, a problem. And there has been no true fault-tolerant solution in the NT cluster market.

Marathon Runs NT With Endurance

Marathon Technologies Corp. of Box-borough, Massachusetts (telephone 800 884-6425) has a unique solution that brings fault tolerance to any Windows NT Server or cluster node. The *Endurance*® 4000 product, now in qualified release, has no single point of failure. It is fault tolerant. By riding through any failure without operator intervention, it then recovers automatically and transparently when the failed component is restored.

Importantly, the NT operating system and all user shrink-wrapped applications are unaware that the *Endurance* 4000 is providing fault tolerance — no special application software versions are required. In fact, Wolfpack or other cluster software can run on Endurance-hardened servers.

Architecture & Operation

Four computers are required to make an *Endurance* 4000 configuration. Two computers are a "mirrored" pair of Computing Elements (CE) running lock-stepped that can be any matched, uni-processor Pentium Pro desktop or server PC. The other two computers are Pentium Pro-class I/O Processors (IOP).

Marathon has tested its product with the leading server hardware platforms such as Compaq, Dell, HP, IBM, and Micron.

The *Endurance 4000* hardware connects a Computing Element to its IOP at the NT hardware abstraction layer (HAL) — that is, under the level of accessibility by normal programs. In effect, NT's processing of services is split between the CE and the IOP. The *Endurance 4000* automatically synchronizes the NT operating system, applications, data storage, and Ethernet interfaces.

More-over, site disaster recovery is built in — each half of the configuration is connected by an optical fiber cable that can be 1.5 kilometers (a mile) from the other half; enough for a building or campus environment. This is an important feature as building fires and floods that disrupt access to facilities are more common than generally known. Thus, the *Endurance 4000* is ideally suited as a fault-tolerant server or as a fault-tolerant cluster node.

It Just Keeps on Ticking ...

Because both halves — one each CE and IOP — are operating in synchronization, no instructions are missed or data lost when either CE or IOP is lost. Users do not see even a hiccup on failure. There is even a port to alert an external alarm system. When the failed electronics, disk, network interface, or power supply is replaced, CEs are synchronized by copying memory, and IOPs are synchronized by automatically copying up-to-the-moment data to the new disk. This is true continuous processing for NT servers and cluster nodes.

Aberdeen Group Conclusions

Users have been asking their NT servers to perform ever-more business-critical application processing. When these application systems fail, business is disrupted and costs and lost opportunities can sky-rocket. And fail they will.

Marathon's *Endurance 4000* is a well-architected, affordable, hardware-based solution that adds almost two orders of magnitude to NT application availability — to about 99.99% application up-time. This is accomplished as if by magic to the shrink-wrapped applications running on otherwise industry-standard Intel hardware platforms.

To us, the benefits of the *Endurance 4000* are obvious and compelling. The \$24,995 price is also reasonable given the value of the applications likely to be protected. If, as we expect, Marathon's *Endurance 4000* breezes through its server hardware compatibility tests, then this valuable new product category — fault-tolerant NT servers and clusters — will have a break-away leader.

— Peter S. Kastner



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